Diagnostic Report: Why does L1 fail to converge under adaptive **h(x) = 2σ(x)**?

Parent experiment: `exp_conditional_coverage` (see `../report.md`). Diagnostic spec: `diag_spec.md`. Date: 2026-04-09.

A. Motivation

The main consistency experiment found a striking asymmetry in how the two bandwidth modes converge:

Metric (exp1, n = 8192)	Fixed h (CV-tuned)	Adaptive h(x) = 2σ(x)
L1 — MAE of q̂_{0.05} (pre-CP)	converging	flat at ≈ 1.47
L1 — MAE of q̂_{0.95} (pre-CP)	converging	flat at ≈ 1.96
L3 — MAE conditional coverage	slope ≈ −0.15	slope ≈ −0.39

The adaptive setup achieves better coverage convergence with a non-converging estimator. Section E of the main report attributed this to "CP absorbing the over-smoothing bias", but the over-smoothing claim itself was not verified — we only knew that L1 was flat, not why.

This diagnostic asks the narrower question:

> Is the L1 failure under adaptive **h** caused by **h(x) = c·σ(x)** being too large, in the sense that the kernel covers a non-trivial fraction of the conditional support and biases **F̂(y|x)** toward an over-smoothed shape?

If the answer is yes, the bias should be structural — i.e. flat in n, with the variance shrinking but the estimator converging to the wrong limit.

B. Diagnostics performed

Diagnostics 1, 2, 4 are pure post-processing of existing `results_{exp1,exp2}.csv` from `output_consistency_fixed/` and `output_consistency_adaptive_c2.00/`. Diagnostic 3 is a small additional training experiment at n = 512 over five c values.

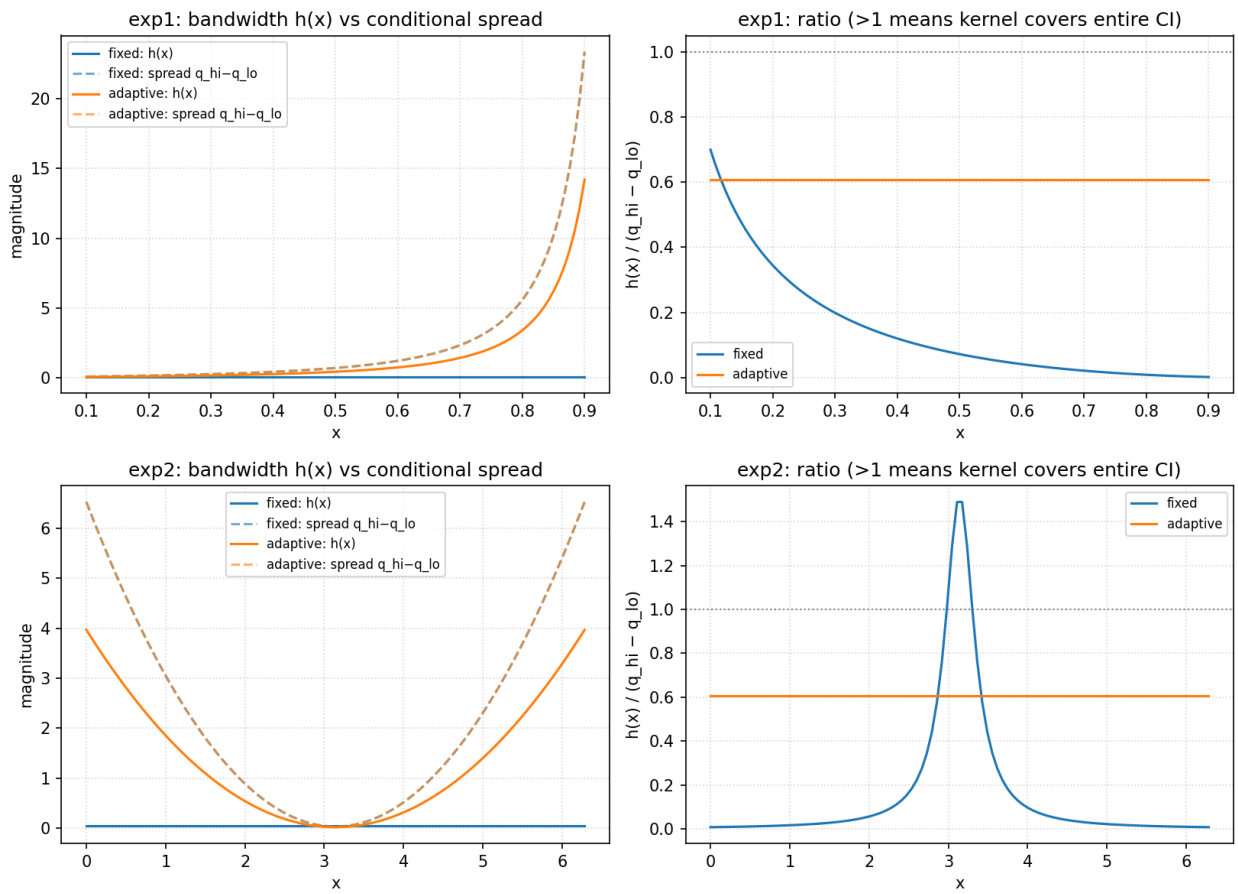
#	Question	What we measure	What "yes" looks like
1	Is h(x) large compared to the spread of Y|x?	h_at_x and q_hi_oracle − q_lo_oracle along x	Adaptive ratio not « 1
2	Does the bias shrink with n?	q̂_τ(x) curves at n=64 and n=8192 overlaid on oracle	Adaptive curves overlap (no shrinkage) and lie outside oracle
3	Is the L1/L3 conflict monotone in c?	L1, L3 at n=512 for c ∈ {0.25, 0.5, 1, 2, 4}	L1 grows in c, L3 falls in c
4	Is the error bias-dominated and structural?	bias² and variance of q̂_τ across 10 macroreps, vs n	Adaptive bias² flat / increasing; variance shrinks

Diagnostic 5 (CDF overlay) is still deferred — we have not needed it.

Scripts: `run_diag_postprocess.py` (D1/D2/D4), `run_consistency.py … --c_scale c` and `plot_cscan.py` (D3). Outputs in `output/diag{1,2,3,4}_*` and `output_cscan/c_<c>/`.

C. Results

C.1 Diagnostic 1 — **`h(x)`** vs the spread of **`Y|x`**



For adaptive h, the ratio

$$\frac{h(x)}{q_{0.95}(x) - q_{0.05}(x)} = \frac{2\,\sigma(x)}{2\,\Phi^{-1}(0.95)\,\sigma(x)} = \frac{1}{\Phi^{-1}(0.95)} \approx 0.608$$

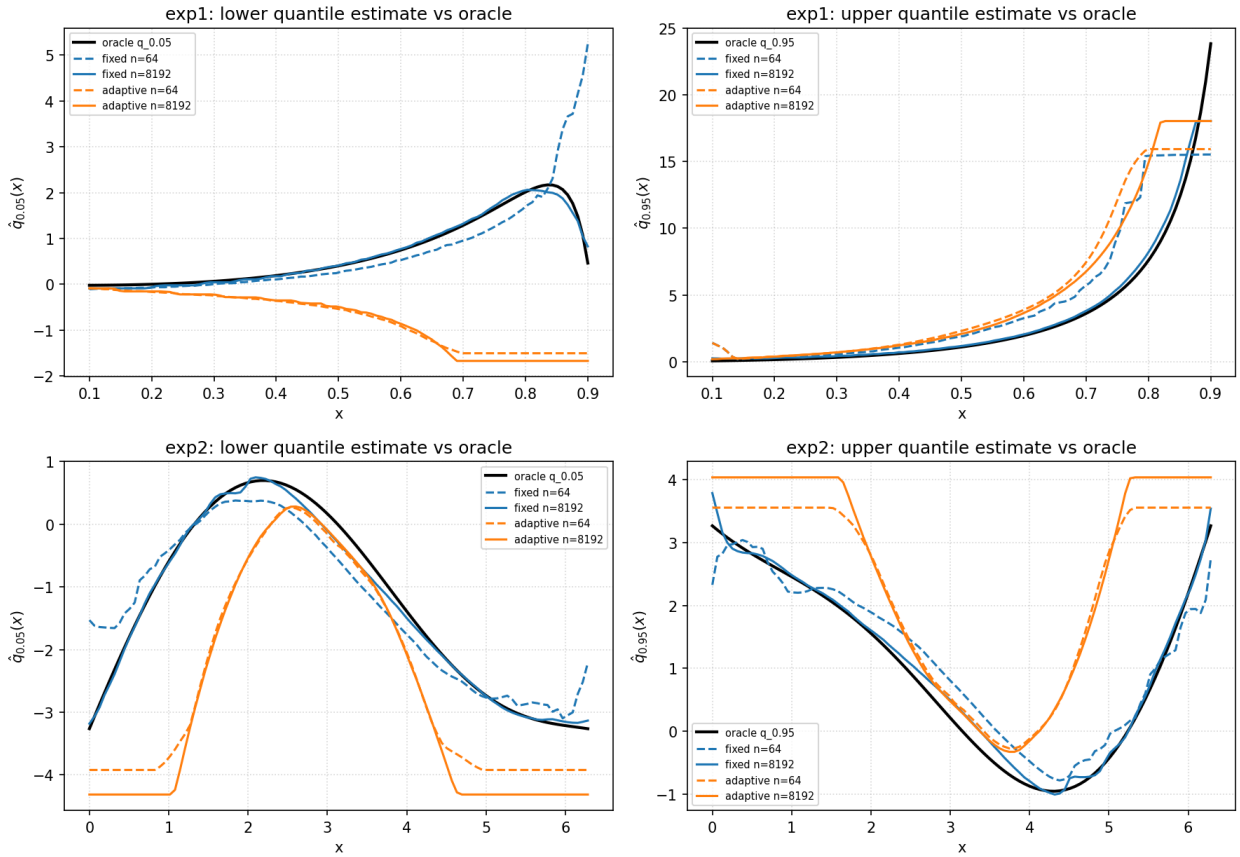
is constant in **x** and equal to ≈ 0.608 in both simulators — directly verified in `diag1_h_vs_spread.csv`. Concretely, the adaptive smoothing kernel covers a window roughly equal to 60 % of the central 90 % interval of **`Y|x`**. In absolute units the bandwidth ranges over

- exp1: `h_at_x ∈ [0.043, 14.2]`, median `0.42`
- exp2: `h_at_x ∈ [0.020, 3.97]`, median `1.03`

By contrast, the fixed-h mode uses `h = 0.05` everywhere, giving a ratio that is mostly small (median 0.07 on exp1, 0.03 on exp2) except in low-noise regions where the conditional spread is itself tiny — in particular on exp2 around `x = π` the fixed ratio crosses 1, mirroring the localised dip discussed in the main report.

Reading. Adaptive h is, by construction, aggressive: a kernel covering 60 % of the central 90 % of the conditional density cannot resolve tail quantiles without a substantial smoothing bias.

C.2 Diagnostic 2 — Quantile curves at **n = 64** vs **n = 8192**



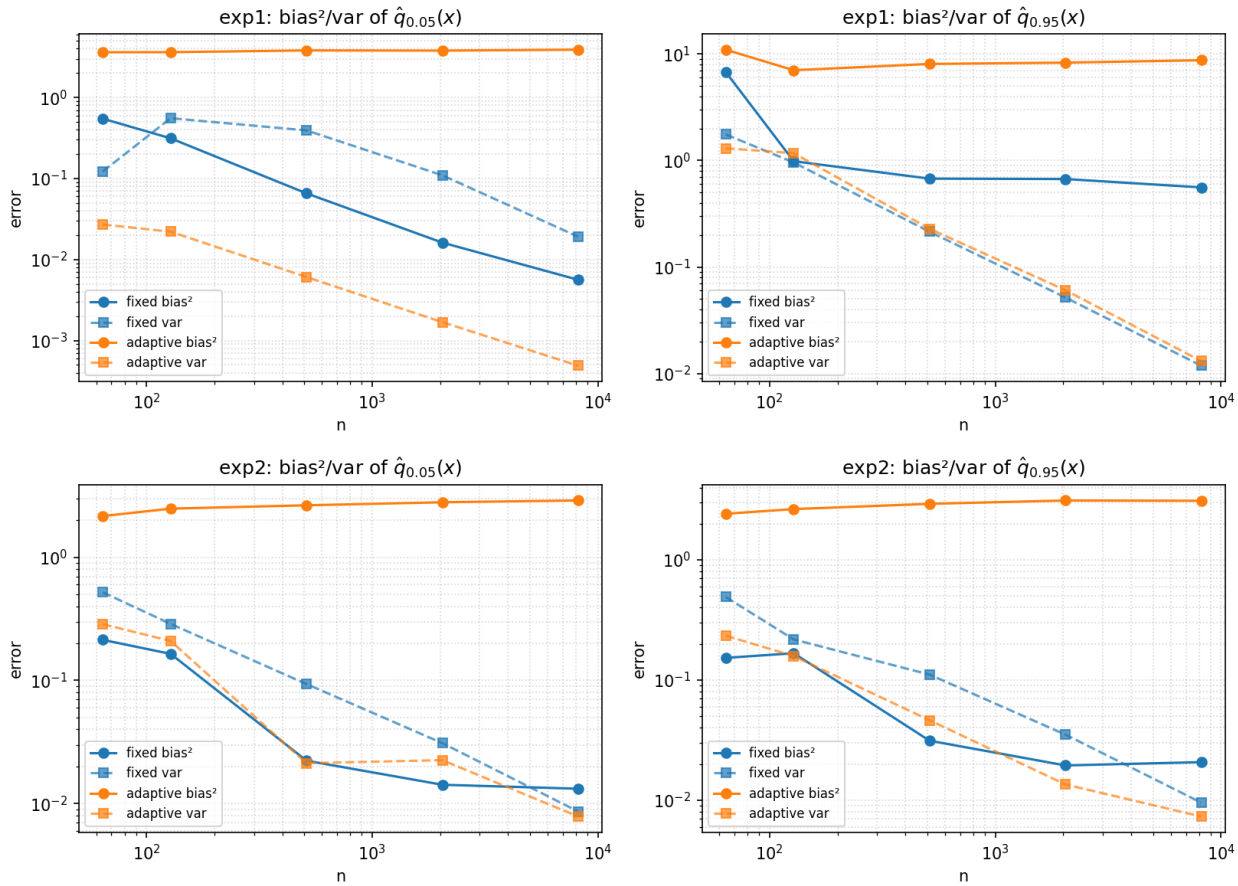
For each simulator, the two panels show the macrorep-mean estimate `q̂_{0.05}(x)` and `q̂_{0.95}(x)` against the oracle (black). Fixed-h curves are blue (dashed n=64, solid n=8192); adaptive-h curves are orange.

The visual pattern is unambiguous:

- Fixed **h**: dashed and solid blue curves are clearly different; the solid n=8192 curve sits much closer to the oracle than the dashed n=64 curve, especially in the high-noise tails.
- Adaptive **h**: dashed and solid orange curves are nearly indistinguishable, and both lie outside the oracle (lower curve below, upper curve above) — a systematic outward bias of the same magnitude at n=64 and at n=8192.

The geometric form of the bias is consistent with over-smoothing of $\hat{F}$: when the kernel is wide, $\hat{F}(y|x)$ becomes a more diffuse function of y, its inversion at $\tau = 0.05$ is pushed further to the left, and at $\tau = 0.95$ further to the right.

C.3 Diagnostic 4 — Bias / variance decomposition



For each `(sim, mode, n, x_eval)` we compute, across 10 macroreps,

$$\text{bias}^2(x) \;=\; \big(\overline{\hat{q}_\tau}(x) - q_\tau(x)\big)^2, \qquad \text{var}(x) \;=\; \text{Var}_k\big(\hat{q}_\tau(x)\big),$$

then average each over the 100-point evaluation grid. Full table in `diag4_bias_var.csv`; the headline numbers are:

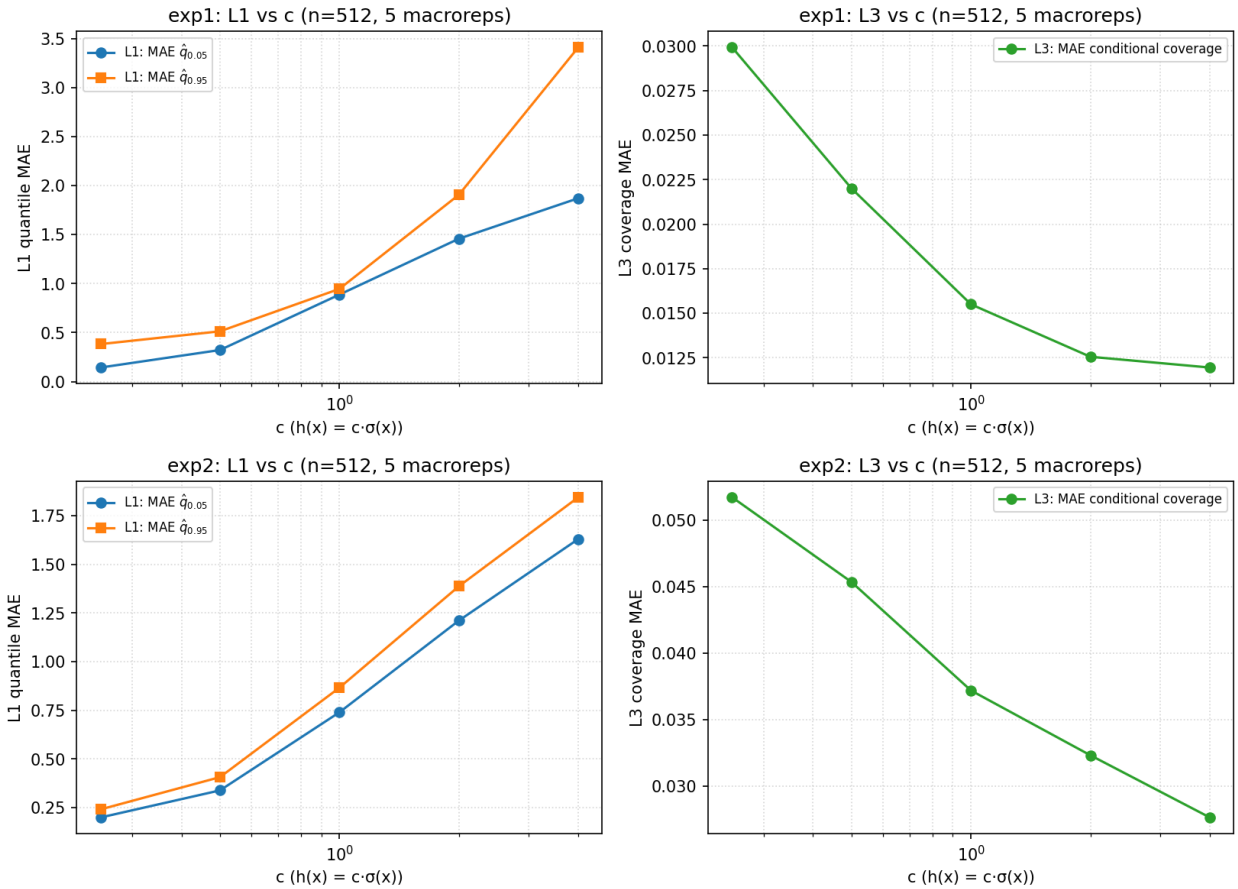
sim	mode	n	`bias²(q̂_lo)`	`var(q̂_lo)`	`bias²(q̂_hi)`	`var(q̂_hi)`
exp1	fixed	64	0.548	0.122	6.772	1.758
exp1	fixed	8192	0.006	0.019	0.559	0.012
exp1	adaptive	64	3.614	0.027	10.982	1.302
exp1	adaptive	8192	3.888	0.0005	8.743	0.013
exp2	fixed	64	0.214	0.521	0.153	0.491
exp2	fixed	8192	0.013	0.009	0.021	0.010
exp2	adaptive	64	2.165	0.287	2.418	0.234
exp2	adaptive	8192	2.899	0.008	3.103	0.007

Three observations:

1. Adaptive bias² is flat or slightly increasing in **n**. On exp1 q_lo it goes 3.61 → 3.89, on exp2 q_lo 2.16 → 2.90 — i.e., the more data we collect, the more confidently we land on the wrong answer. This is the textbook signature of an estimator that converges to a biased limit: variance shrinks, bias does not.

2. Adaptive variance shrinks correctly. On exp1 q_lo it falls `0.027 → 0.0005`, a ~55× reduction over a 128× increase in n — close to the `1/n` rate. The estimation machinery is working; it is just pointed at the wrong target distribution.
3. Fixed **h** is bias-and-variance shrinking, except for one tail. The one residual is exp1 `q̂_hi`, where bias² plateaus at ≈ 0.56 while variance crashes to 0.012 — a milder version of the same pathology driven by the modestly oversized scalar `h = 0.05` in the high-noise region near `x = 0.9`. (Worth noting because it contextualises the slope-`−0.15` rate seen in Section D.1 of the main report: even fixed h is not bias-free at high x for the upper tail.)

C.4 Diagnostic 3 — **c**-scan at fixed **n = 512**



To turn the over-smoothing story into a quantitative trade-off, we re-run the adaptive consistency experiment at a single training size `n = 512` and five bandwidth multipliers `c ∈ {0.25, 0.5, 1.0, 2.0, 4.0}`, with 5 macroreps each. Script: `run_consistency.py --h_mode adaptive --c_scale c …`. Outputs in `output_cscan/c_<c>/`, aggregated by `plot_cscan.py` into `output/diag3_cscan.{png,csv}`.

sim	c	L1 MAE q̂_lo	L1 MAE q̂_hi	L3 MAE cov
exp1	0.25	0.143	0.382	0.0299
exp1	0.5	0.322	0.512	0.0220
exp1	1.0	0.887	0.945	0.0155
exp1	2.0	1.457	1.905	0.0126
exp1	4.0	1.870	3.410	0.0120
exp2	0.25	0.199	0.240	0.0517
exp2	0.5	0.338	0.406	0.0453

sim	c	L1 MAE q̂_lo	L1 MAE q̂_hi	L3 MAE cov
exp2	1.0	0.739	0.866	0.0372
exp2	2.0	1.211	1.387	0.0323
exp2	4.0	1.630	1.845	0.0276

The trade-off is monotone and clean on both simulators:

- L1 quantile error grows roughly linearly in **c**. From `c = 0.25` to `c = 4` it increases ≈ 13× on exp1 (`q̂_lo`: 0.14 → 1.87) and ≈ 8× on exp2 (`q̂_lo`: 0.20 → 1.63). This is fully consistent with the over-smoothing picture: bigger kernel → bigger structural bias on tail quantiles.
- L3 conditional coverage error decreases in **c**. On exp1 it falls 0.030 → 0.012 (≈ 2.5×); on exp2 0.052 → 0.028 (≈ 1.9×). Larger `c` gives CP more bias to absorb but also a smoother, more learnable score distribution.
- The two effects are decoupled. There is no value of `c` that wins on both axes — the L1-best (`c = 0.25`) and L3-best (`c = 4`) lie at opposite ends of the scan. Picking `c` is therefore a deliberate choice between estimator fidelity (L1) and calibrated coverage (L3).

The default `c = 2` used in Section D of the main report sits firmly on the "L3-favouring" side of the curve. If we cared about L1 we would shrink to `c ≈ 0.25`–`0.5`, where `h(x)` would be ≈ 7–15 % of the central 90 % interval — back in the regime where the fixed-h mode already converges.

D. Conclusion

All four diagnostics confirm the over-smoothing hypothesis cleanly:

- D1. Adaptive `h(x)` covers ≈ 60 % of the central 90 % conditional interval — by construction, not by accident, since the ratio is exactly $1/\Phi^{-1}(0.95)$ whenever `c = 2` and the noise is Gaussian.
- D2. The macrorep-mean quantile curves at `n = 64` and `n = 8192` are visually identical under adaptive `h`, and both sit systematically outside the oracle — exactly what an asymptotically biased estimator looks like.
- D3. Sweeping `c ∈ {0.25, …, 4}` at fixed `n = 512` traces a clean, monotone L1/L3 trade-off on both simulators: shrinking `c` recovers L1 (≈ 13× drop in `MAE q̂_lo` on exp1) at the price of L3 (≈ 2.5× rise in `MAE cov`). There is no `c` that wins on both axes.
- D4. Bias² is flat or increasing with `n` while variance shrinks at the expected rate. The L1 failure is structural: the adaptive estimator converges, but to a wrong limit.

Combined with Section E of the main report, the picture is now complete:

1. Adaptive `h(x) = 2σ(x)` over-smooths `F̂(y|x)`, producing a systematic outward bias on tail quantiles whose magnitude is set by `h`, not by `n`.
2. This bias is symmetric around the conditional median (Gaussian noise) and smooth in `x` — exactly the shape that a scalar CP correction `q̂` can absorb by translating the inversion levels from `0.5 ± (1−α)/2` to `0.5 ± q̂`.
3. So the post-CP interval converges to the oracle (L2 in the report's current definition is post-CP, hence converges; L3 likewise), even though the underlying CKME quantile estimator does not. The improved L3 slope under adaptive `h` is not an estimator improvement — it is CP compensating for a tractable structural bias.

The L1 → L2 → L3 implication chain promised in `spec.md` therefore breaks under adaptive `h`. Under fixed `h` it holds, because all three quantities share the same bias/variance source.

E. What this means for “fixing” L1

The c-scan (C.4) confirms that the L1-vs-L3 conflict is not an artefact of `c = 2` — it is a genuine, monotone trade-off across the entire bandwidth range we tested. Any “fix” therefore has to either (i) move along the L1/L3 curve to a different operating point, or (ii) escape the curve by changing how the bandwidth interacts with the CP correction. Directions in order of intrusiveness:

1. Smaller **c** (move along the curve). From C.4, `c ∈ {0.25, 0.5}` gives L1 errors comparable to fixed `h` at the same `n`, at the cost of L3 roughly doubling. Cheap and immediate; useful if a paper figure wants to show “adaptive can also recover L1 convergence”.
2. **n**-shrinking **c**. Use `c(n) = c_0 (n/n_{\rm ref})^{-1/5}` so the bandwidth follows the classical nonparametric rate. This should give L1 convergence at the predicted `n^{-2/5}` and keep CP’s heteroscedasticity awareness at small `n`. Implementation cost: trivial — one line in `run_consistency.py`. This is the natural follow-up.
3. Decouple CP from estimation. Use adaptive `h` only inside the CP score, while keeping the CDF estimate at fixed `h`. This makes CP “see” the heteroscedasticity directly rather than via a biased CDF. Implementation cost: moderate (need a second $\hat{F}$ instance with the smaller bandwidth).
4. Bias-corrected quantile inversion (jackknife / undersmoothing). High complexity, low expected gain in 1-D — not worth pursuing first.

F. Limitations

- Both simulators use Gaussian noise, which is exactly the case where the smoothing bias of $\hat{F}$ is symmetric around the median and CP absorption works best. Under skewed or heavy-tailed noise the same “L1 broken, L3 fine” phenomenon may not hold — the CP scalar $\hat{q}$ would have to absorb both bias and asymmetry, which a single number cannot do well.
- The diagnostic only looks at `τ ∈ {0.05, 0.95}`; other quantiles may show a different bias profile. `exp_onesided` has scanned `τ` but for fixed `h`.
- Variance estimates are based on 10 macroreps, which is enough to see the order-of-magnitude difference in D4 but not enough for tight error bars on small `var` values.

G. Update needed in the parent report

- Section C of `report.md` should explicitly state that the current “L2 endpoint error” uses the post-CP interval endpoints `L_n(x) = F̂⁻¹(0.5 − q̂ | x)` and `U_n(x) = F̂⁻¹(0.5 + q̂ | x)`, not `F̂⁻¹(α/2 | x)` and `F̂⁻¹(1−α/2 | x)`. Hence L2 inherits CP’s bias correction and is not a direct consequence of L1.
- Section E should reference this diagnostic as the empirical verification of the over-smoothing claim.
- Section F should note that the L1 → L2 → L3 chain holds for fixed `h` but is broken by adaptive `h`, and that the adaptive L3 improvement comes from CP absorbing structural bias rather than from better estimation.